Week 6 – Product Review and Rating Management System.

1. Introduction

The Product Review and Rating Management System is used to store and manage customer feedback about products. Customers can give ratings from 1 to 5 stars and write comments about their experience. This system helps identify product quality and customer satisfaction.

2.Review Table Design

The review table stores customer ratings, feedback comments, and review dates. Each review is identified by a unique review ID. The table connects customer and product records using foreign keys.

3. Database & Review Table Creation

The database stores customer, product, and review details. The review table stores the customer ID, product ID, rating, review text, and review date. It helps manage product reviews efficiently.

Output:

Field	Type	Null	Key	Default	Extra
review_id	int	NO	PRI	NULL	auto_increment
customer_id	int	NO	MUL	NULL	
product_id	int	NO	MUL	NULL	
rating	int	NO		NULL	
review_text	varchar(500)	NO		NULL	
review_date	date	NO		NULL	

4. Primary Key, Foreign Key & Other Constraints

A primary key uniquely identifies each review record. Foreign keys connect the review table with the customer and product tables. The not null and check constraints ensure valid review data and ratings between 1 and 5.

Output:

	Field	Type	Null	Key	Default	Extra
▶	review_id	int	NO	PRI	NULL	auto_increment
	customer_id	int	NO	MUL	NULL	
	product_id	int	NO	MUL	NULL	
	rating	int	NO		NULL	
	review_text	varchar(500)	NO		NULL	
	review_date	date	NO		NULL	

5. Sample Data – INSERT Queries

Sample data is inserted into the review table to test the database. Each review contains a valid customer ID, product ID, rating, review text, and date. The IDs must already exist in the customer and product tables.

Output:

	customer_id	customer_name
▶	101	ravi
	102	arun
	103	kumar
	104	raja
	105	john

	product_id	product_name	category_id	price
▶	101	laptop	1	45000.00
	102	mobile	1	18000.00
	103	headphones	2	1500.00
	104	keyboard	2	800.00
	105	mouse	2	500.00

	review_id	customer_id	product_id	rating	review_text	review_date
*	NULL	NULL	NULL	NULL	NULL	NULL

6. Product Review CRUD Queries

CRUD stands for Create, Read, Update, and Delete. These operations help insert new reviews, display existing reviews, modify review details, and remove unwanted reviews.

Output:

	product_name	rating	review_text

7. Review Update and Delete Queries

The update operation changes an existing review's rating or feedback. The delete operation removes a review from the table. These operations help maintain accurate and updated customer feedback.

6. Product Rating Analysis & Report Queries

Rating analysis helps evaluate product quality using customer feedback. It calculates average ratings and review counts for each product. These reports help identify highly rated products and products that need improvement.

7.Conclusion

The Product Review and Rating Management System was successfully created using MySQL. Customer reviews and product ratings were stored and managed using SQL queries. CRUD operations helped to insert, view, update, and delete review records. Rating analysis helped identify highly rated products and products that need improvement. Thus, this project helps to understand customer feedback and improve product quality.